

DARSH SONI

darsh.codes@gmail.com | +91 6376973874 | <https://darsh.is-a.dev> | github.com/darshsoni04 | [linkedin.com/in/darshcmd](https://www.linkedin.com/in/darshcmd)

EDUCATION

Vellore Institute of Technology, Chennai

B.Tech in Computer Science and Engineering (AI & Robotics)

Jul 2024 - Jul 2028

CGPA: 8.38/10

EXPERIENCE

Centre of Cyber Physical Systems, VIT Chennai

May 2026 - Aug 2026

AI Research Intern

- Constructed a spatiotemporal glacier surface velocity forecasting framework for the Dagaard-Jensen Glacier tongue using multi-temporal Sentinel-1 SAR observations (2017- 2025).
- Engineered an end-to-end preprocessing workflow leveraging ESA SNAP offset tracking, geocoding, and spatial masking with a Persistence-Gated Residual Swin architecture, achieving an RMSE of 0.568 m/day, R^2 of 0.856, and Pearson correlation of 0.928.

BSERC, IIT Madras

Jan 2026 - Mar 2026

Space Technology Intern

[github](#)

- Worked on drone systems, PCB prototyping, GIS workflows, and remote-sensing applications for aerospace research.
- Explored CanSat and CubeSat systems, telemetry, and rocketry concepts and applied them to OpenVSP based aircraft modelling of NASA Cessna 210 aircraft model and technical documentation.

RESEARCH & ENGINEERING PROJECTS

AI-Based Railway Crack & Obstacle Detection System | Patent

[youtube](#) | [github](#)

- Designed a railway safety prototype combining MPU6050, IR, and ultrasonic sensors for real-time crack, obstacle, and vibration detection.
- Developed computer-vision pipelines using OpenCV and YOLO for railway crack detection and obstacle-type classification., contributing to a filed patent application.

Multi-Stage Natural Language to ER Diagram & SQL Generation System | Research

[link](#) | [github](#)

- Developed an agentic AI system using Google ADK to transform natural-language database requirements into ER diagrams and SQL schemas.
- Deployed the system as a live web application, combining multi-agent orchestration with automated database schema generation.

Personalized Braille Learning System

[link](#) | [github](#)

- Designed an AI-driven tailored Braille educational platform combining tactile feedback, voice interaction, user profiling, and dynamic practice routines.
- Engineered a customized learning mechanism that assesses student performance, tracks progress metrics, and automatically generates targeted practice sessions.

Gesture-Controlled 5-DOF Robotic Arm using Computer Vision

[github](#)

- Constructed a real-time gesture-driven 5-DOF robotic manipulator using OpenCV visual tracking (HSV color segmentation, contour processing, and kinematic joint calculation) from camera streams.
- Programmed Arduino microcontrollers with a PCA9685 I2C driver for servo actuation and designed a structured binary communication protocol with filtering to eliminate motor chatter.

TECHNICAL SKILLS

- Languages:** C++ | C | Python | Java | JavaScript | R | x86 Assembly | MATLAB
- Computer Science:** Data Structures & Algorithms | Competitive Programming | System Design | Operating Systems | Computer Architecture | Computer Networks | Concurrency
- Systems & Embedded:** Linux | Firmware Development | Microcontroller Programming | Embedded C | Real-Time Systems | Sensor & Peripheral Interfaces
- Robotics:** Manipulator Arms | Humanoid Robotics | Multi-DOF Systems | Motion Control | Inverse Kinematics | Sensor Fusion | Computer Vision | CAD
- AI/ML:** PyTorch | Deep Learning | Computer Vision | YOLO | NLP | Agentic AI | RAG | Time-Series Forecasting
- Development & Data:** Git | Docker | REST APIs | SvelteKit | React | Node.js | Oracle SQL | MongoDB | AWS

ACHIEVEMENTS & LEADERSHIP

- Working on Agile PiperX and LIMO Cobot research at CCPS, research focuses on multi-robot morphing independent interaction.
- Founder of Scientific Paper Society, VIT Chennai; Team Lead, French Club, Animation Club, GDG VIT Chennai.
- Google Student Ambassador, sole student liaison for 34,000+ students at VIT Chennai; conducted GenAI workshops.
- 3rd Place - TechnoVIT CloudOps Hackathon; 2nd Place - MediCode Hackathon; Finalist in 5 other national-level hackathons.
- Top 1% - GirlScript Summer of Code 2025, with 24+ merged pull requests across open-source projects.
- Google Gemini Certified Educator; Deloitte Solutions Architecture and Data Analytics Certified.